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Bromide and lithium transport in soils under long-term cultivation of alfalfa and wheat

G. Yousefi[a], A. Safadoust[a,*], A.A. Mahboubi[b], B. Gharabaghi[b], M.R. Mosaddeghi[c], B. Ahrens[d], H. Shirani[e]

[a] Department of Soil Science, College of Agriculture, Bu-Ali Sina University, Hamadan, Iran
[b] School of Engineering, University of Guelph, Guelph, Ontario, Canada
[c] Department of Soil Science, College of Agriculture, Isfahan University of Technology, Isfahan, Iran
[d] Department of Geography, University of Guelph, Guelph, Ontario, Canada
[e] Department of Soil Science, Vail-E-Asr University of Rafsanjan, Rafsanjan, Iran

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ABSTRACT

The combined effects of two soil textures and two types of crop management were investigated using lithium and bromide tracers transport under saturated flow conditions. Leaching tests were carried out through intact columns of two soils, a clay loam (CL) and sandy loam (SL), each cropped with either wheat (*Triticum aestivam*) (W) or alfalfa (*Medicago sativa*) (A) for 11 years. A saturated steady state flow condition was established using tap water prior to injecting a pulse of 0.005 M LiBr solution onto the surface of the soil columns. Breakthrough curves (BTCs) for leached Br^- and Li^+ in the soil columns (especially under alfalfa cultivation) exhibited an early arrival time and greater concentrations, indicating preferential flow effects. Relative 5% arrival times were 0.07, 0.11, 0.24 and 0.31 for CL-A, SL-A, CL-W, and SL-W, respectively; its smaller values confirmed the higher possibility of preferential flow under alfalfa than under wheat. The Br^- concentration was higher than the Li^+ concentration. With the exception of soils under alfalfa (CL-A and SL-A), the peak of the BTCs for Br^- occurred earlier than that for Li^+, by about 0.4 and 1.2 pore volumes for the CL-W and SL-W cases, respectively. Clay loam soil under alfalfa showed higher Br^- and Li^+ concentration levels when compared to sandy loam soil under alfalfa crop production. In the soils under alfalfa cultivation, structural cracks, root channels, and earthworm burrows were the cause of higher leached concentrations for both tracers when compared to the soil under wheat. Therefore, alfalfa-induced changes in soil structure lead to continuous macropores, a result of the decomposition of penetrating roots. Our results show that agricultural-management practices (i.e. the type of cropping) can play an important role in making groundwater vulnerable to leached solutes.

1. Introduction

The contamination of water resources by various chemicals is of significant concern for water quality, motivating the need for research into the transport of these chemicals in soils (Caron et al., 1996; Gish et al., 1991a; Edwards et al., 1997; Mohanty et al., 1998). Factors such as rapid preferential flow and the transport of solutes through large pores and cracks in substrates have been linked to high concentrations of solute contaminants in groundwater (Topp and Davice, 1981; Kanwar, 1991). Studies have shown that both soil texture and long term tillage and crop management practices have a strong influence on soil properties such as pore volume, pore-size distribution, and pore-network (Mahboubi et al., 1993; Lal and Van Doren, 1990; Lal et al., 1990, 1994). Aside from the accepted benefits of conservation tillage (decreasing soil erosion and reducing pollution of surface water), researchers have shown that conservation tillage leads to increased leaching of chemicals to groundwater resources (Edwards et al., 1997; Steenhuis et al., 1990b; Gish et al., 1991b). Lal and Van Doren (1990) found that in the soils of central Ohio, no-till plots had higher infiltration rates than tilled plots. Mahboubi et al. (1993) reported that hydraulic conductivity was significantly higher in no-tillage treatments than both moldboard plowing and chisel plowing treatments.

Other important factors that can significantly modify soil structural properties include differences in antecedent soil properties,

* Corresponding author. Tel.: +98 811 4424366; fax: +98 811 4424012.
E-mail addresses: safadoust@basu.ac.ir, safadoust@gmail.com, asafadoust@yahoo.com (A. Safadoust).

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climate conditions, soil texture, history of cultural management and wheel tracks (Mahboubi et al., 1993; Schwartz et al., 2003). It was found by Koestel et al. (2011) that preferential flow is directly related to the soils with more than 8% clay content. It has been reported that soil hydraulic properties, soil structure, pore volume, pore-size distribution, and bulk density are affected by land use and agricultural management practices (Afyuni et al., 1994; Caron et al., 1996; Edwards et al., 1993; Hu et al., 2009). Bormann and Klaassen (2008) noted that soil structure and the geometry and volume of pores can vary over time and are strongly influenced by soil biological activity and the land use.

Numerous studies have shown that the infiltration capacity and leaching behavior of solutes are strongly affected by the type of soil and its hydraulic properties (Knappe et al., 2002; van Es et al., 2006). Zhou et al. (2008) reported that differences in soil hydraulic properties were impacted more by the type of vegetation than by the type of soil. Soil pore networks are believed to be improved by both meadow and leguminous cover crops (Lal et al., 1990, 1994). The strong taproot of alfalfa has also been reported to increase the hydraulic conductivity of soils under or close to saturated conditions (Caron et al., 1996). It was concluded by Caron et al. (1996) that a change from corn to alfalfa significantly impacted hydraulic conductivity and thereby solute transport relative to lands cropped with corn. Perfect et al. (1990) and Chantigny et al. (1997) reported improved soil hydraulic properties attributed to land use and land cover. Alfalfa cultivation has also been reported to significantly increase soil hydraulic conductivity relative to soils that are under corn cultivation (Li and Ghodrati, 1994). Alfalfa, as a perennial and long-lived plant, constructs larger and deeper root systems than wheat as an annual and short-lived plant (Huggins et al., 2001); thereby, preferential flow paths could be created by decaying alfalfa root system as a result of increased total porosity and continuous macropores (Rasse and Smucker, 1998). Gibbs and Reid (1988) reported that the large numbers of pores formed by root were classified as macropores.

As indicated above, most studies have evaluated how various factors affect soil hydraulic properties near the soil surface or the leaching of nutrients (Bormann and Klaassen, 2008; Bachmair et al., 2009). However, knowledge about their indirect influence on water flow, groundwater recharge, and solute transport is lacking. The question remains whether these changes near the soil surface, resulting from different land use and land cover, have any net effect on groundwater recharge over long time periods or whether they just impact soil structure. Particularly for semiarid region agricultural soils, it is important that we understand the influence of land use (tilled or no tilled) and different plant covers, not only on the development of soil structure and soil hydraulic properties, but also on water flow and solute transport. As mentioned above, structural heterogeneities, such as changes in soils due to vegetation and treatment, can result in water flux heterogeneities and/or preferential flow.

The objective of this study is to provide a better understanding of tracers transport through undisturbed soil columns collected from cultivated soils in natural conditions under land cover as well as land use effects on soil saturated hydraulic properties over long time periods. Soil hydraulic properties, solute transport and leached concentrations were measured to develop an understanding of differences between different land covers (alfalfa, and wheat) and common annually tillage implantation in wheat cultivation and no-tilled condition in alfalfa cultivation. Since there is a demand for a better understanding of the environmental and management controlling factors on solute leaching, the present study assesses the impact of (i) two contrasting land use systems (alfalfa and wheat) and (ii) two texturally different soil (clay loam and sandy loam) on the leaching of two conservative and non-conservative tracers (Br^- and Li^+).

2. Materials and methods

2.1. The study site and soil properties

The soils were collected from the 0 to 30 cm horizon of two sites of sandy loam (SL) and clay loam (CL) soils that were classified as Typic Xerorfluvents and Typic Haploxerepts, respectively, according to USDA classification (Soil Survey Staff, 2010). These soils were collected from the province of Hamadan in western Iran during the early fall in 2008. Each soil had been under cultivation of either wheat, *Triticum aestivam* L. (Gramineae) (conventionally tilled) or alfalfa, *Medicago sativa* L. (Legume) for 11 years. It should be noted that the samples were taken from a small flat plot (50 cm × 50 cm) with minimum sampling space. The plots were under the same plant density and no visible crack and fracture was observed in the plots.

Selected physical and chemical properties of the soil are given in Tables 1 and 2, respectively. The soil pH was measured in water (ratio of soil:water 1:2.5) (Jackson, 1967). Soil electrical conductivity (EC) was determined using an EC-meter in 1:5 soil:water suspension (Rhoades, 1996). Calcium carbonate ($CaCO_3$) content of soil was measured using the back-titration method (Sims, 1996). Total organic carbon was determined using the wet-digestion method (Walkly and Black, 1934). Cation exchange capacity (CEC) was measured using 1 mol L^{-1} ammonium acetate (pH = 7 NH_4OAC) method (Kacar, 1995).

The particle size distribution was determined using the hydrometer method (Bouyoucos, 1962). The particle density was measured by the pycnometer method (Gee and Bauder, 1986). Bulk density was measured on the undisturbed soil cores with 5 cm diameter and 7.5 cm height (Klute, 1986). Total porosity was calculated from bulk density (BD) and particle density (PD), (TP = 1 − BD/PD). Pore size distribution was determined from the water retention curve obtained with the pressure plate method. The relative volume of the pores with effective diameters greater than 30 μm was defined as macropores (Macro-P) (Danielson and Sutherland, 1986). Mean weight diameter (MWD) of soil aggregates was determined by the wet-sieving method (Yoder, 1936). Saturated hydraulic conductivity (K_s) of the soil was measured by the constant-head procedure (Klute and Dirksen, 1986).

2.2. Soil column samplings procedure

The laboratory experiments were conducted on undisturbed soil columns that had two different textures (sandy loam, SL and clay loam, CL) and were cultivated with two different cropping practices (wheat, W and alfalfa, A) under saturated flow condition. It is known that root morphologies and physiologies of alfalfa and wheat significantly differ, as a consequence the movement of water and solutes through the soils could be considerably different. Alfalfa has an extensive taproot which may penetrate the soil 7–9 m. It is not unusual, however, for the root system to be highly branched whereas the mature root system of wheat ordinarily reaches a maximum depth of 30–150 cm (Meek et al., 1990; Mitchell et al., 1995; Caron et al., 1996; Aggarwal et al., 2006).

The undisturbed soil column samples were taken using galvanized cylinders with a 3 mm wall thickness, a 16 cm inner diameter, and a 32 cm length. The inner wall of the cylinders was moistened by liquid paraffin to prevent the possibility of flow along the soil column interface and to facilitate the insertion of the cylinder into the soil. In total, four column types were sampled: (1) sandy loam columns under wheat cultivation (SL-W); (2) sandy loam columns under alfalfa cultivation (SL-A); (3) clay loam columns under wheat cultivation (CL-W); and (4) clay loam columns under alfalfa cultivation (CL-A). All columns were sampled in four replicates for a total of sixteen soil columns.

Table 1
Chemical properties of the studied soils under different crops (average for 0–30 cm).[a]

Soil-crop combination[b]	EC ($dS\,m^{-1}$)	pH	OC ($g\,kg^{-1}$)	$CaCO_3$ ($g\,kg^{-1}$)	CEC ($cmol\,kg^{-1}$)
CL-A	0.30	7.3	10.5	120	17.2
CL-W	0.24	7.5	9.3	110	18.4
SL-A	0.28	7.2	9.1	50	11.9
SL-W	0.14	7.6	5.2	45	11.7

[a] EC, Electrical conductivity; pH, Acidity; OC, Organic carbon content; $CaCO_3$, Calcium carbonate content; CEC, Cation exchange capacity.
[b] CL and SL represent clay loam and sandy loam soils, respectively; A and W stand for alfalfa and wheat, respectively.

Table 2
Physical properties of the studied soils under different crops (average for 0–30 cm).[a]

Soil-crop combination[b]	Sand ($g\,100\,g^{-1}$)	Silt ($g\,100\,g^{-1}$)	Clay ($g\,100\,g^{-1}$)	BD ($Mg\,m^{-3}$)	PD ($Mg\,m^{-3}$)	TP ($cm^3\,cm^{-3}$)	Macro-P ($cm^3\,cm^{-3}$)	MWD (mm)
CL-A	41.2 (±12.0)[c]	22.0 (±6.0)	47.20 (±5.0)	1.37 (±0.10)	2.61 (±0.1)	47.20 (±11.5)	25.70 (±1.7)	3.52 (±0.25)
CL-W	39.0 (±8.0)	22.3 (±10.0)	46.30 (±15.0)	1.40 (±0.11)	2.61 (±0.0)	46.30 (±5.7)	14.80 (±4.6)	2.31 (±0.11)
SL-A	69.3 (±18.0)	12.7 (±3.0)	45.30 (±11.0)	1.45 (±0.10)	2.67 (±0.1)	45.30 (±18.0)	19.60 (±8.3)	3.22 (±1.0)
SL-W	71.2 (±21.0)	10.8 (±5.0)	43.10 (±18.0)	1.54 (±0.12)	2.72 (±0.2)	43.10 (±13.2)	13.120 (±1.5)	0.54 (±0.06)

[a] BD, Bulk density; PD, Particle density; TP, Total porosity, Macro-P, Macroporosity; MWD, Mean weight diameter of soil aggregates.
[b] CL and SL represent clay loam and sandy loam soils, respectively (USDA classification); A and W stand for alfalfa and wheat, respectively.
[c] Figures in the parentheses are the standard deviations.

2.3. Batch adsorption tests

Batch experiments were carried out for each soil to determine Li^+ adsorption (K_d) using adsorption isotherms. After homogenization, 10 g of sieved soil (>2 mm) was equilibrated with 10 mL aqueous solutions ($CaCl_2$) containing 0, 10, 20, 50, 100, 250, 500, 1000, and 1200 mg L^{-1} Li^+, separately. The suspension was gently hand-mixed, left for 24 h at room temperature, filtered, and then analyzed. The amount of adsorbed Li^+ was calculated from the difference between the initial and equilibrium concentrations of Li^+. Results were fitted to Freundlich isotherms (Fouqué-Brouard and Fournier, 1996; Benoit et al., 1999; Nemeth-Konda et al., 2002; Akhtar et al., 2003; Moradi et al., 2005) expressed as:

$$S = K_d C^{\beta} \quad (1)$$

where S is adsorbed Li^+ concentration [$M\,L^{-3}$], C is concentration of Li^+ in soil solution [$M\,L^{-3}$], K_d [$M\,L^{-3}$]$^{-\beta}$ and β [−] are empirical constants. Three replicates were conducted for each treatment.

2.4. Leaching experiments

This study investigated the effects of soil texture and cropping practices on the transport of two chemical tracers, bromide (Br^-) and lithium (Li^+), under saturated, steady-state flow conditions. Each soil column was saturated with tap water from the base for two days and then placed on a funnel to measure the saturated hydraulic conductivity by the constant-head method (Klute, 1986) (Fig. 1). It is necessary to note that Li^+ has been used as a cationic tracer by several researchers (Claridge et al., 1999; Akhtar et al., 2003; Karen et al., 2005), which can be attracted to negatively-charged soil particles such as clay and organic matter particles.

In relation to the dispersing effect of Lithium on clay aggregates, we did not find any significant dispersion effects on soil aggregates for the applied solution (containing 399 mg L^{-1} of Br and 34.7 mg L^{-1} of Li, equivalent LiBr powder) in the prior experiment. Furthermore we did not find any significant clay or particles in the outlet flow or decreeing of flow rate during leaching experiments.

The characteristics of the established saturated steady-state flow of the soil columns under different treatments are summarized in Table 3.

To create similar flow rates on all columns for each treatment, a constant flux equal to the highest K_s of each treatment was applied on the corresponding columns (i.e. 4.44 cm h^{-1} for SL-W, 7.93 cm h^{-1} for SL-A, 7.51 cm h^{-1} for CL-W, 8.38 cm h^{-1} for CL-A). Once the steady-state flow was established, a pulse input of 0.005 M LiBr solution (C_0) was applied on the surface of soil columns with the same rate of applied flow flux followed by steady-state leaching for 4 pore volumes. The mean time (averaged over all soil textures and croppings) of pulse input was 48 min for all of the treatments.

The soil solution (10 mL) was sampled at a depth of 25 cm, at 0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1, 1.2, 1.4, 1.6, 1.8, 2, 2.2, 2.4, 2.6, 2.8, 3, 3.2, 3.4, 3.6, 3.8 and 4 pore volumes (PVs) after applying the LiBr pulse. Each soil solution effluent sample was tested for its concentration (C) of Br^- (M) and Li^+ (M). The Br^- concentration was determined using a bromide selective electrode (Metrohm AN-S-217) while

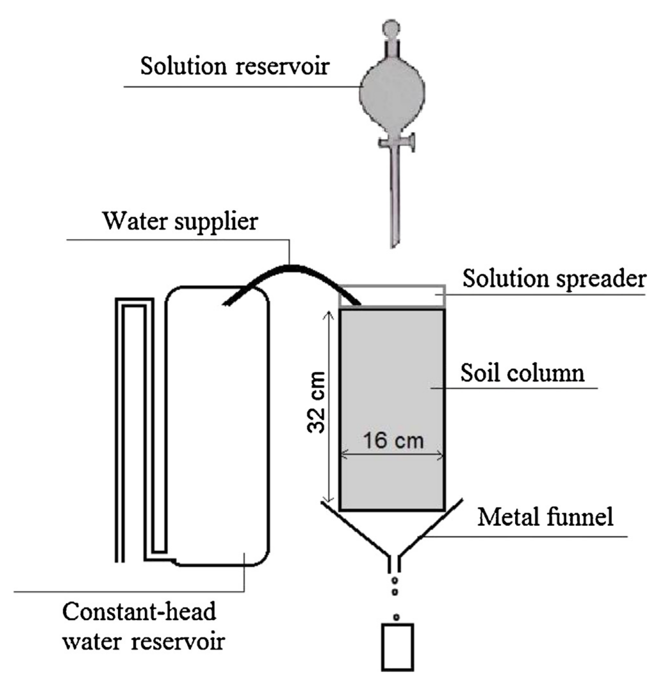

Fig. 1. Schematic diagram of leaching experiment.

Table 3
Characteristic of the established saturated flow condition.[a]

Soil-crop combination[b]	θ_s ($cm^3\ cm^{-3}$)	q ($cm\ h^{-1}$)	$\overline{\upsilon}$ ($cm\ h^{-1}$)	PV (cm^3)
CL-A	0.472 (±0.09)[c]	8.38 (±1.09)	17.8 (±2.34)	2372 (±133)
CL-W	0.463 (±0.01)	7.51 (±1.00)	16.2 (±2.58)	2330 (±52)
SL-A	0.451 (±0.03)	7.93 (±2.30)	17.5 (±3.66)	2283 (±71)
SL-W	0.432 (±0.05)	4.44 (±0.24)	10.2 (±0.99)	2169 (±102)

[a] θ_s, Saturated water content; q, Steady-state flux rate; $\overline{\upsilon}$, Apparent pore water velocity; PV, Pore volume.
[b] CL and SL represent clay loam and sandy loam soils, respectively; A and W stand for alfalfa and wheat, respectively.
[c] Figures in the parentheses are the standard deviations.

the Li^+ concentration was determined using a flame photometer (Jenway PFP7).

Recovery of Br^- and Li^+ in the leachate was quantified by integrating the surface under influent curve and the breakthrough/effluent curve (BTCs). The relative solute mass in the effluent to that in the influent was used as an index of recovery. The presence of preferential flow was also quantified by relative 5% arrival time (t_{r5}) obtained from breakthrough curve (BTC) experiments. Relative 5% arrival time is the ratio between the time when 5% of the solute reaches the outlet and the mean arrival time of the solute (Koestel et al., 2011). Smaller values of relative 5% arrival times is positively correlated to the strength of preferential flow, where the larger one implies the weak preferential flow.

2.5. Statistical analysis

An analysis of variance ($p < 0.05$) was performed using the SAS package (SAS Institute, 1990) to determine the significant effects of treatments on some soil physical properties, K_d for Li^+, solute recovery in the effluent and relative 5% arrival time (t_{r5}). The impacts of soil texture (CL and SL) and cropping types (W and A) were assessed in a randomized factorial design.

3. Results and discussion

3.1. Effects of soil texture and cropping type on soil physical properties

Physical properties of the studied soils are summarized in Table 2. Results demonstrated the significant effects of soil texture and cropping type on the physical properties. The improvement in structure, indicated by a large increase in MWD, was most pronounced for CL-A with 41.2% clay and 1.05% organic matter (Tables 1 and 2). In these columns, there was a 1.5-fold increase in the MWD relative to CL-W (3.52 mm *vs.* 2.31 mm) and a 6-fold increase was observed for SL-A (3.22 mm) compared to SL-W (0.54 mm). Many studies have linked the effects of cropping systems on soil physical properties to changes in soil organic matter (SOM) (Dormaar, 1983; Haynes, 2000; Larney et al., 1997) due to particular soil management scenarios (Christensen, 1992). Angers et al. (1999) reported that annual crops often decrease the stability of soil aggregates.

The calculated pore volume (PV) was slightly higher in the CL-A, when compared to the other treatments (Table 3). This suggests that in CL-A more pores were generated and connected, and may participate in the transport of water and solutes. As expected, the flux density was higher in CL-A than in the others (Table 3), suggesting that there was not only higher porosity, but also more consecutiveness between the pores (Table 2). Our findings are in agreement with Ehlers and Claupein (1994), who reported greater macroporosity and pore continuity under conservation agriculture due to increased biological activity and soil swelling and shrinking.

The K_s increased 12% in the CL-A (8.38 $cm\ h^{-1}$) over CL-W (7.51 $cm\ h^{-1}$) columns and 79% in the SL-A (7.93 $cm\ h^{-1}$) over SL-W columns (4.44 $cm\ h^{-1}$) (Table 2). Caron et al. (1996) also reported that four-year old alfalfa resulted in a difference in K_s in comparison to corn. The formation of macropores, due to decomposition of roots in cropped soils (Barley, 1954), wetting and drying cycles, freezing and thawing in clayey soils (Safadoust et al., 2011, 2012a), and earthworms (Safadoust et al., 2012b) was reported.

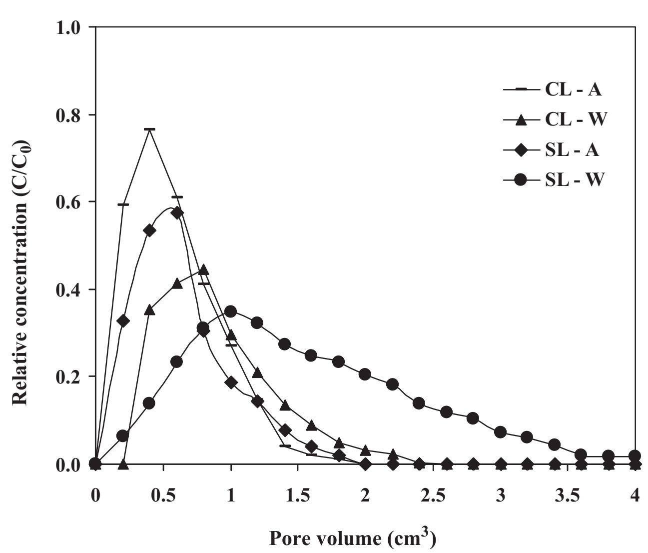

Fig. 2. Measured breakthrough curves of bromide under experimental treatments (average of replicates). CL and SL represent clay loam and sandy loam soils, respectively; A and W represent alfalfa and wheat, respectively.

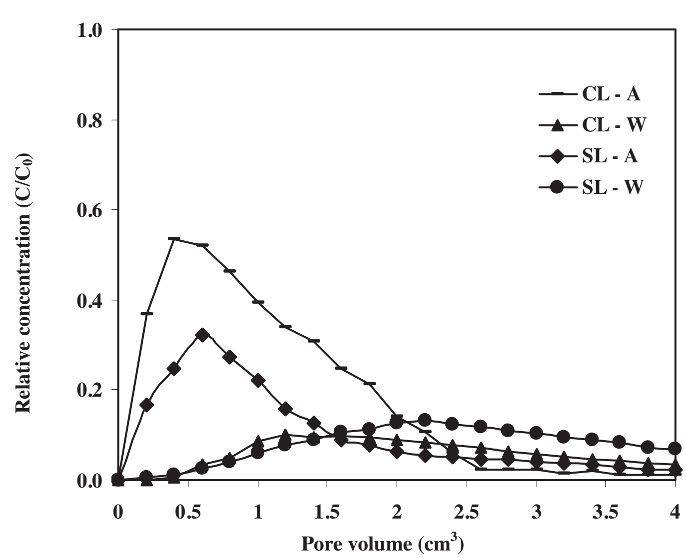

Fig. 3. Measured breakthrough curves of lithium under experimental treatments (average of replicates); CL and SL represent clay loam and sandy loam soils, respectively; A and W represent alfalfa and wheat, respectively.

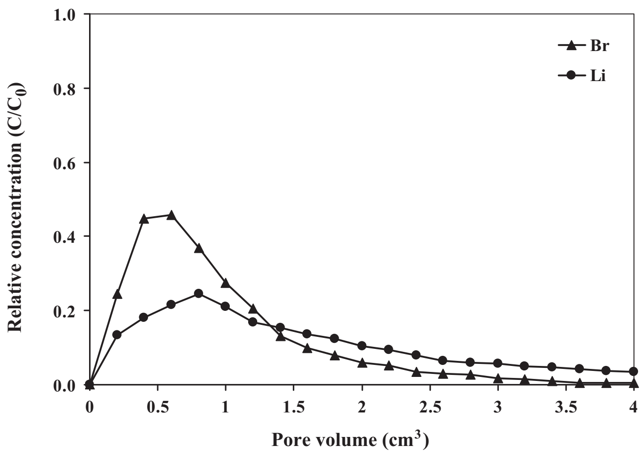

Fig. 4. Breakthrough curves of bromide (Br^-) and lithium (Li^+) (averaged over different treatments).

Although infiltration rate can be limited in the active root-soil system, decomposition of roots could result in the creation of water-conducting channels or macropores (Barley, 1954).

3.2. Transport of bromide and lithium

The breakthrough curves (BTCs) of Br^- and Li^+ are shown in Figs. 2 and 3, respectively. Both soil texture and cropping affected the transport of Br^- and Li^+. The figures show that both tracers appeared in effluent shortly after application of the solution, especially under alfalfa cultivation. The rapid movement of tracers through preferential pathways (mainly in the clay loam column) caused early tracer arrivals visible in the BTCs. This movement could be linked to the decaying of roots, which allows water to flow rapidly through the macropores (Meek et al., 1989, 1990; Murphy et al., 1993).

In general, maximum Br^- and Li^+ concentrations in effluent were observed in the CL-A columns because of higher K_s (Fig. 2). Likely, the greater amount of continuous pores facilitated the transport of tracers. This is especially likely for Li^+, regardless of the larger apparent adsorptive capacity (i.e. clay) in these columns. In reality, the occurrences of preferential pathways are more evident in clayey soils. Continuity of macroporosity was well documented as a main reason of tracer transport in structured soils (Akhtar et al., 2003; Mosaddeghi et al., 2009; Safadoust et al., 2012a). The measured concentration of Li^+ in effluent was lower and delayed in comparison with Br^- during leaching experiments (Fig. 4), a result of its adsorbable property.

The comparison of the Br^- and Li^+ BTCs (Fig. 2) indicates that in the CL-A and SL-A columns, Br^- was observed in the effluent within 0.02 and 0.12 PV while Li^+ was not observed in the effluent until 0.21 and .0.25 PV respectively, after the initiation of leaching. Very little Br^- was observed in the leachate of SL-W columns, even after 0.52 PV (Fig. 2).

The BTCs of alfalfa were different from wheat, with more preferential flow for the alfalfa (Fig. 5a2 and b2). Other studies also reported the presence of preferential flow under alfalfa due to rapid

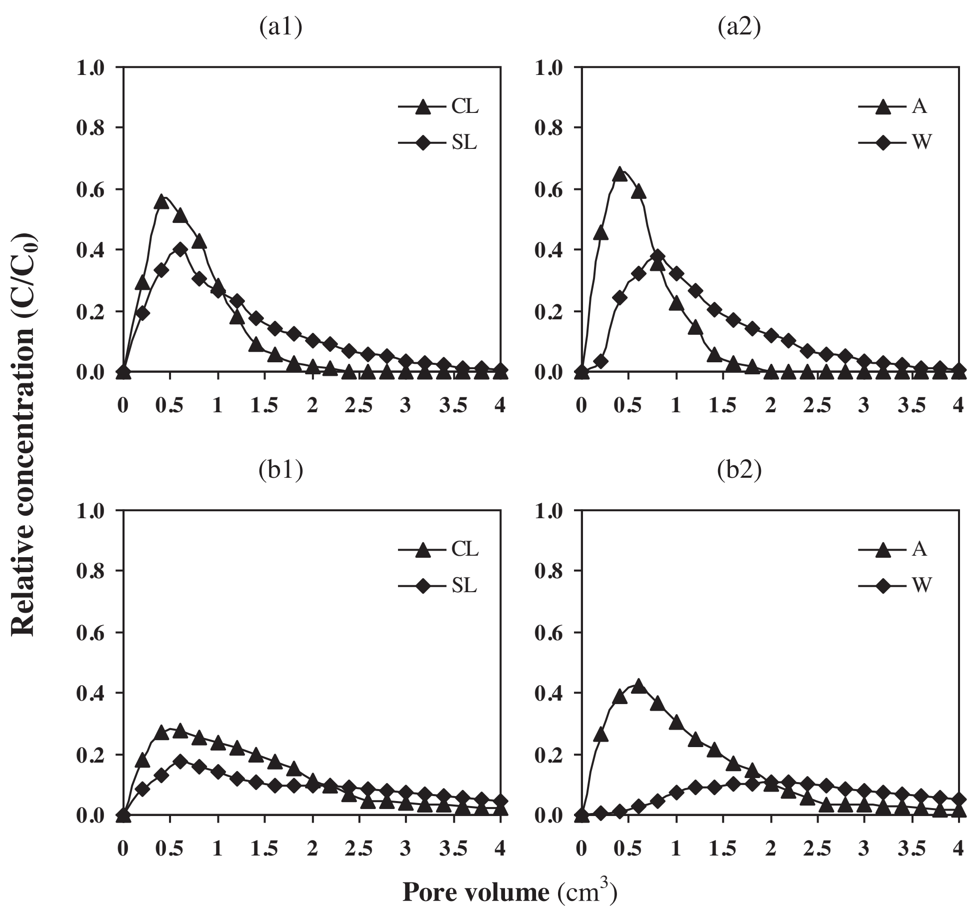

Fig. 5. The breakthrough curves of bromide (a) and lithium (b) as affected by soil texture (1) and cropping (2). CL and SL represent clay loam and sandy loam soils, respectively; A and W represent alfalfa and wheat, respectively.

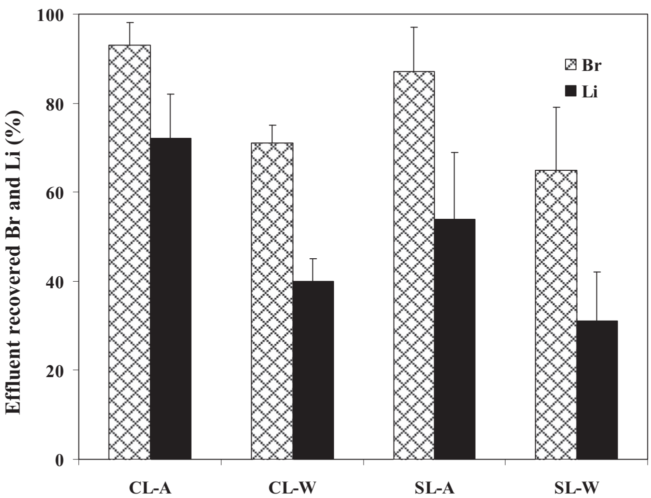

Fig. 6. Total recovery rates of bromide (Br^-) and lithium (Li^+) as affected by soil-crop combinations. The error bars show the standard deviations of the three replicates. CL and SL represent clay loam and sandy loam soils, respectively; A and W stand for alfalfa and wheat, respectively.

flow velocity in the decayed roots pathways of this plant (Meek et al., 1989, 1990; Murphy et al., 1993).

Relative concentrations of Br^- and Li^+ in effluent decreased in SL columns (Fig. 5a1 and b1). This trend was observed for all treatments regardless of cropping. The decrease in Br^- and Li^+ concentrations and the delayed appearance of solutes in the leachate of SL columns shows that aggregation, structural stability, and continuity of the macropores that facilitated contamination transport are not enhanced in coarse-textured soils with low clay content. Saturated and near-saturated hydraulic conductivities of clayey soils can increase markedly in time, due to the formation of cracks resulting from shrinking and swelling (Jabro, 1996; Azevedo et al., 1998). The higher levels of contamination in leachate of fine-textured soils compared to coarse-textured soils have also been previously reported by several researchers (Mosaddeghi et al., 2009; Safadoust et al., 2011, 2012a; Unc and Goss, 2003).

The significant differences between the mean values of recovered Br^- and Li^+ in various treatments (Fig. 6) confirms the higher possibility of the presence of preferential paths under alfalfa than under wheat, as well as in CL columns when compared to SL columns.

Scott et al. (1994) demonstrated that soil dynamic properties such as bulk density, pore size distribution, and hydraulic conductivity vary temporally, a result of imposed management practices, such as cultivation. The Br^- recovered (as % of applied) in leachate, decreased from 93% to 71% in the CL-A *vs.* CL-W and from 87% to 65% in the SL-A *vs.* SL-W (Fig. 6). Trapping of bromide in the very fine pores or in the death-end pores by diffusion into the immobile phases were reported by several researchers as a description for uncovered bromide (Jiang et al., 2005; Dousset et al., 2007; Safadoust et al., 2011, 2012a).

The amount of Li^+ recovered in drained water, for CL-A and CL-W was 72% and 40% of the applied Li^+ mass respectively; corresponding values for SL-A and SL-W were 54% and 31% (Fig. 6). It is likely that the higher water velocity in the cracks and macropores of soils under alfalfa limited the Li^+ adsorption in the fine pores. Although many studies have reported the limited risk of water contamination of unstructured clay and clay loam soils as a result of their lower permeability (Ou et al., 1999; Safadoust et al., 2011), the presence of macropore have clearly been demonstrated to increase hydraulic conductivity, consequently increasing the risk of contamination in those soils (Topp and Davis, 1981; Kissel et al., 1973). Gibbs and Reid (1988), Wang et al. (1985) and Meek et al. (1989, 1990) have also noted that changes in the cropping system can lead to changes in soil macroporosity and hydraulic properties. Mitchell et al. (1995) reported that the temporal variation of hydraulic conductivities resulted from the activity and type of root systems. They associated the rapid water velocity of a cultivated clayey soil to decayed roots of perennial plants, which can generate more stable biopores when compared to annual plants.

Table 4
Freundlich isotherm parameters obtained by the batch experiment.[a]

Soil-crop combination[b]	K_d ($mg^{(1-\beta)}$ $(cm^3)^\beta$ g^{-1})	β –	R^2
CL-A	23.55 (±1.42)[c]	0.98 (±0.02)	0.982
SL-A	17.81 (±3.81)	1.14 (±0.14)	0.991
CL-W	28.36 (±2.09)	1.00 (±0.08)	0.973
SL-W	15.42 (±3.73)	0.74 (±0.03)	0.985

[a] K_d and β are empirical constants of Freundlich isotherm parameters.
[b] CL and SL represent clay loam and sandy loam soils, respectively; A and W stand for alfalfa and wheat, respectively.
[c] Figures in the parentheses are the standard deviation.

The presence of preferential flow can also be specified by the relative 5% arrival time (t_{r5}) obtained from the BTCs. The t_{r5} stands for the relative time at which 5% of solutes pass the porous media; its smaller value indicates earlier appearance of solute in the leachate and a greater extent of preferential flow. The calculated t_{r5} values were significantly affected by soil-crop combination effects ($p < 0.01$). The t_{r5} values were 0.07, 0.11, 0.24 and 0.31 for CL-A, SL-A, CL-W and SL-W treatments, respectively, which confirmed the higher possibility of preferential flow in clay loam columns and under alfalfa cultivation in comparison with sandy loam columns and those under wheat cultivation. In the other words, the soils under alfalfa showed stronger preferential flow than the soils under wheat.

Table 4 shows the adsorption constants (K_d and β) of Li^+ batch adsorption experiments of the Freundlich isotherm Eq. (1) for different treatments. The obtained K_d values were 23.6, 17.8, 28.4 and 15.4 $mg^{(1-\beta)}(cm^3)^\beta g^{-1}$ for CL-A, SL-A, CL-W and SL-W treatments respectively, with β values ranging from 0.74 to 1.14. The greater adsorption capacity associated with greater amount of silt and clay, pH, and CEC (see Tables 1 and 2) presumably resulted in greater value of K_d in the CL-A treatment. Akhtar et al. (2003) also reported greater K_d in fine-textured soil (greater silt and clay content). Adsorption constants (Table 4) and recovered lithium (Fig. 6) indicates the importance of soil structure in lithium transport in comparison with soil chemical properties.

4. Conclusions

1. Our results show that the extent of contamination leaching was significantly affected not only by soil texture/type but also with a greater extent by cropping history. Despite the possible higher adsorption of Li^+ in clay loam soil particles than in sandy loam soil, continuity of the macropores in clay loam soil may be the reason for higher recovery of Li^+ in CL-A than SL-A columns.
2. For the considered treatments, flux density (q), saturated water content (θ_s), macroporosity (Macro-P), mean weight diameter (MWD) and the calculated relative 5% arrival time (t_{r5}) increased in the following order: CL-A < SL-A < SL-W < CL-W.
3. The smaller values of t_{r5} may be strongly correlated to early appearance, relatively asymmetrical and left-shifted shape of BTCs and also can be linked to the presence of non-equilibrium transport processes in soils under alfalfa than in soil under wheat cultivation. The increased groundwater vulnerability under alfalfa can be attributed to more macropores and

channels created by decaying roots leading to greater soil hydraulic conductivity.

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